Ramp Enterprise Campus Network — Technical Documentation

Oct 18, 2026 · @Syed Jafri

1. Overview

This project is a five-device Cisco campus design for a fictional mid-size Ramp office: two Layer 3 distribution switches, two Layer 2 access switches and one internet edge router, serving HR, Finance, Engineering, Guest and management segments. It is built to run in a lab (Packet Tracer, GNS3 or EVE-NG) and demonstrates VLAN/VTP, Rapid-PVST, HSRP, ACLs, NAT/PAT and subnetting.

This document is the complete technical reference for the design as committed in syedjafri06193/Cisco-Enterprise-Network-Arch-proj (release v1.0). Every value below is taken from the device configs; where the configs and the existing markdown notes disagree, the configs win and the gap is called out in section 14.

Design goals

- Segment users by business function (HR, Finance, Engineering, Guest)
- Give every VLAN a redundant default gateway (HSRP across DIST1/DIST2)
- Prevent Layer 2 loops and control the forwarding path (Rapid-PVST, root at distribution)
- Route between VLANs at the distribution layer (SVIs)
- Enforce policy with ACLs: Guest is internet-only, Finance cannot reach Engineering
- Provide internet access through a single PAT edge
- Ship a reproducible diagram, IP plan and validation checklist

Out of scope in v1.0

DHCP, DNS, wireless controllers, dynamic routing, device hardening (SSH, AAA, logging, NTP), QoS and monitoring are not configured. Section 14 lists what a production rollout would add.

Repository layout

Path	Contents
`README.md`	Project summary, VLAN and IP tables, policy highlights
`configs/DIST1.txt`, `configs/DIST2.txt`	Distribution switch configs: VLANs, SVIs, HSRP, STP root, ACLs, uplinks
`configs/ACCESS1.txt`, `configs/ACCESS2.txt`	Access switch configs: trunks, user ports, management SVI
`configs/EDGE1.txt`	Edge router: inside/outside interfaces, static routes, PAT
`diagrams/ramp-enterprise-network.drawio`	draw.io topology diagram
`docs/*.md`	Short design notes (business context, VLAN/VTP, IP plan, STP/HSRP, ACLs, NAT, validation, change log)

2. Business requirements

The office needs four user segments with different trust levels, plus a management network, and every segment must keep its gateway if one distribution switch fails.

Segment	VLAN	Users	Trust level	Required access
HR	10	People Ops workstations, HR systems access	Internal	Baseline: any internal and internet (to be tightened later)
Finance	20	Accounting operations, sensitive workflows	Internal, restricted	Internet and internal, but never Engineering
Engineering	30	Developer workstations, internal tooling	Internal, broad	Broad east-west access and internet

Segment	VLAN	Users	Trust level	Required access
Guest	40	Visitor and contractor Wi-Fi / BYOD	Untrusted	Internet only, no internal resources
Management	99	Switch management interfaces	Infrastructure	Reachable by network operators

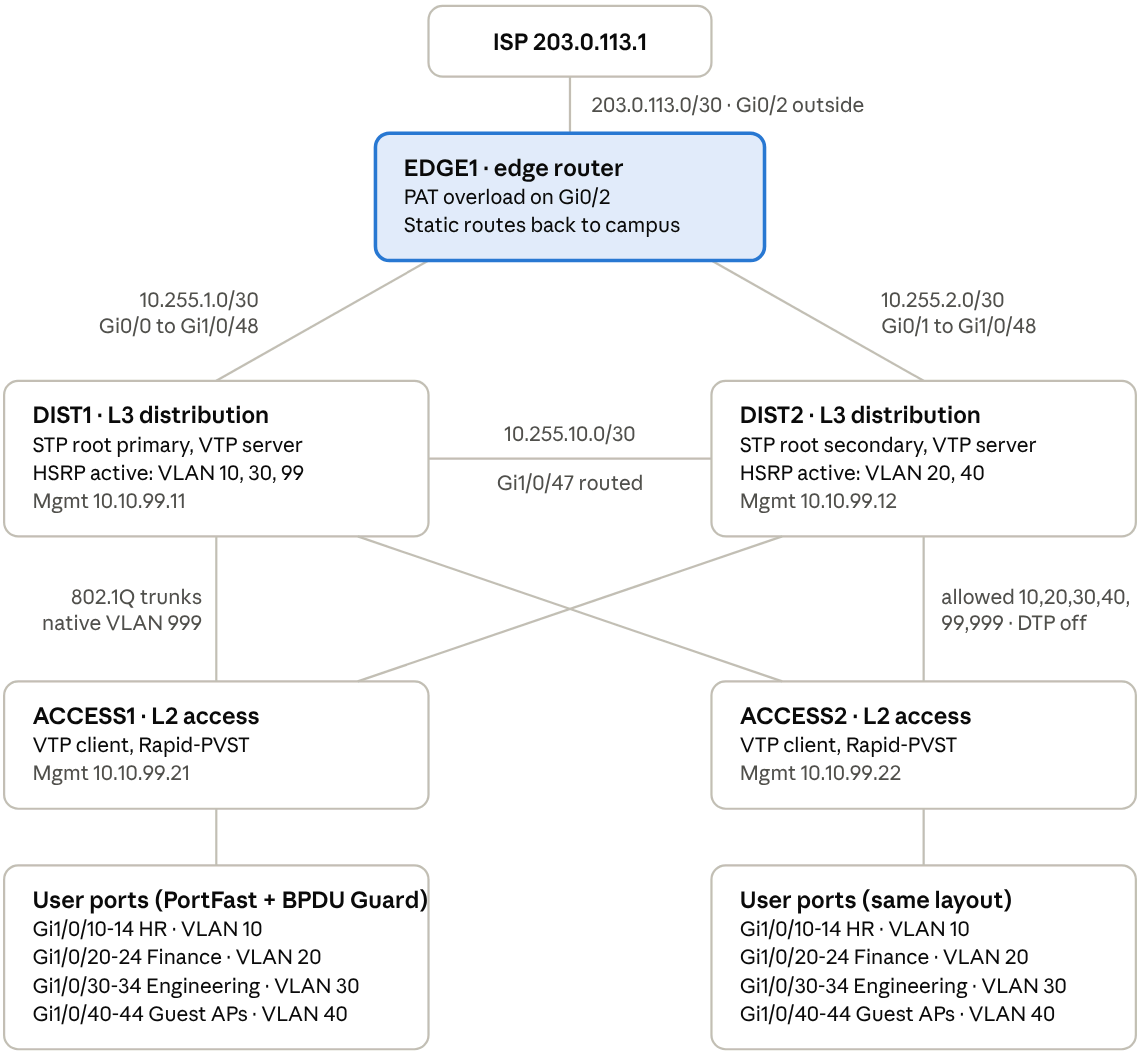
Network priorities

1. High availability for user default gateways
2. Secure isolation of Guest traffic
3. Restricted access between Finance and Engineering
4. Centralised routing at distribution with predictable Layer 2 behaviour
5. Simple, testable internet access through NAT at the edge

3. Topology and physical design

The network follows a collapsed two-tier campus model with a separate internet edge: each access switch has one trunk to each distribution switch, and each distribution switch has its own routed uplink to EDGE1.

Every access switch dual-homes to both distribution switches



campus topology · 5 devices, 3 tiers

The four trunks form a single Layer 2 loop (DIST1 – ACCESS1 – DIST2 – ACCESS2), which Rapid-PVST breaks by blocking one port per VLAN. The DIST1–DIST2 link is a routed /30, so there is no direct Layer 2 path between the distribution switches; HSRP hellos travel through the access switches.

Device roles

Device	Role	Platform assumed	Key functions
EDGE1	Internet edge router	ISR-class router (Gi0/0–0/2)	PAT, default route to ISP, static routes to campus VLANs

Device	Role	Platform assumed	Key functions
DIST1	Distribution, L3 switch	Catalyst 3650/3850-class (Gi1/0/x)	VTP server, STP root primary, SVIs, HSRP, ACLs
DIST2	Distribution, L3 switch	Same as DIST1	VTP server, STP root secondary, SVIs, HSRP, ACLs
ACCESS1	Access, L2 switch	Catalyst-class (Gi1/0/x)	VTP client, user ports, management SVI
ACCESS2	Access, L2 switch	Same as ACCESS1	VTP client, user ports, management SVI

Link table

Link	Side A	Side B	Type	Subnet / VLANs
ISP uplink	EDGE1 Gi0/2	ISP	Routed, NAT outside	203.0.113.0/30
Edge transit 1	EDGE1 Gi0/0	DIST1 Gi1/0/48	Routed, NAT inside	10.255.1.0/30
Edge transit 2	EDGE1 Gi0/1	DIST2 Gi1/0/48	Routed, NAT inside	10.255.2.0/30
Inter-distribution	DIST1 Gi1/0/47	DIST2 Gi1/0/47	Routed	10.255.10.0/30
Trunk	DIST1 Gi1/0/1	ACCESS1 Gi1/0/1	802.1Q trunk	10,20,30,40,99,999 (native 999)
Trunk	DIST1 Gi1/0/2	ACCESS2 Gi1/0/1	802.1Q trunk	10,20,30,40,99,999 (native 999)
Trunk	DIST2 Gi1/0/1	ACCESS1 Gi1/0/2	802.1Q trunk	10,20,30,40,99,999 (native 999)
Trunk	DIST2 Gi1/0/2	ACCESS2 Gi1/0/2	802.1Q trunk	10,20,30,40,99,999 (native 999)

The access-side port pairing is inferred from interface descriptions ("TRUNK TO DIST1" on Gi1/0/1, "TRUNK TO DIST2" on Gi1/0/2). The draw.io file in `diagrams/` shows the same five devices and VLAN list.

4. VLAN, VTP and trunking

Six VLANs are defined once on the distribution switches and propagated to the access layer by VTP version 2 in domain `RAMP-CAMPUS`; all trunks are static 802.1Q with DTP disabled and an unused native VLAN.

VLAN database

VLAN	Name	Purpose	SVI / gateway	ACL on SVI (inbound)
10	RAMP-HR	People Ops / HR workstations	10.10.10.1 (HSRP)	HR_IN
20	RAMP-FIN	Finance / Accounting endpoints	10.10.20.1 (HSRP)	FIN_IN
30	RAMP-ENG	Engineering workstations	10.10.30.1 (HSRP)	none
40	RAMP-GUEST	Guest Wi-Fi / contractor BYOD	10.10.40.1 (HSRP)	GUEST_IN
99	RAMP-MGMT	Network device management	10.10.99.1 (HSRP)	none
999	NATIVE-DROP	Native VLAN on trunks; carries no user traffic	none	n/a

VTP

Setting	Value
Domain	`RAMP-CAMPUS`
Version	2
Password	Configured (stored in plain text in the repo; see 14)

Setting	Value
Servers	DIST1, DIST2
Clients	ACCESS1, ACCESS2

Both distribution switches are VTP servers and carry identical `vlan` blocks, so either can add or rename VLANs. The access switches define no VLANs locally; they learn the database over the trunks. Because VTP v2 lets any server with a higher configuration revision overwrite the domain, VLAN changes should be made on one server only (DIST1 by convention).

Trunk standard

Every trunk (eight ports in total) uses this template:

```
interface GigabitEthernet1/0/x
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40,99,999
 switchport nonegotiate
```

- **Static trunking:** `switchport mode trunk` plus `nonegotiate` disables DTP, so no port can be talked into trunking.
- **Native VLAN 999:** untagged frames land in a VLAN with no SVI and no access ports, which defeats VLAN double-tagging attacks against user VLANs.
- **Allowed list:** only the six defined VLANs cross the trunks, which limits broadcast scope and keeps unknown VLANs off the wire.
- **Encapsulation command:** `switchport trunk encapsulation dot1q` is required on older Catalyst models and Packet Tracer 3560/3650 images; on 2960 or Catalyst 9000 platforms it is rejected and can be omitted.

Access port standard

```
interface range GigabitEthernet1/0/10 - 1/0/14
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
```

The same pattern repeats for VLAN 20 (Gi1/0/20–24), VLAN 30 (Gi1/0/30–34) and VLAN 40 (Gi1/0/40–44, guest access points) on both access switches.

5. IP addressing plan

All campus space comes from 10.10.0.0/16, with the third octet equal to the VLAN ID; transit links use /30s from 10.255.0.0/16 and the ISP link uses documentation space 203.0.113.0/30.

Addressing conventions

- `.1` = HSRP virtual IP (the gateway hosts use)
- `.2` = DIST1 SVI, `.3` = DIST2 SVI (user VLANs 10–40)
- `.11` / `.12` = DIST1/DIST2 in VLAN 99, `.21` / `.22` = ACCESS1/ACCESS2 in VLAN 99
- Third octet = VLAN ID, so an address identifies its segment at a glance
- Transit /30s: 10.255.<n>.0, lower usable address on the distribution side for edge links

Subnets

Subnet	VLAN / link	Mask	Usable hosts	Gateway	Usable range for hosts
10.10.10.0/24	VLAN 10 HR	255.255.255.0	254	10.10.10.1	10.10.10.4 – .254
10.10.20.0/24	VLAN 20 Finance	255.255.255.0	254	10.10.20.1	10.10.20.4 – .254
10.10.30.0/24	VLAN 30 Engineering	255.255.255.0	254	10.10.30.1	10.10.30.4 – .254
10.10.40.0/24	VLAN 40 Guest	255.255.255.0	254	10.10.40.1	10.10.40.4 – .254
10.10.99.0/24	VLAN 99 Management	255.255.255.0	254	10.10.99.1	10.10.99.23 – .254 (after devices)
10.255.1.0/30	DIST1 – EDGE1	255.255.255.252	2	n/a	.1 DIST1, .2 EDGE1
10.255.2.0/30	DIST2 – EDGE1	255.255.255.252	2	n/a	.1 DIST2, .2 EDGE1
10.255.10.0/30	DIST1 – DIST2	255.255.255.252	2	n/a	.1 DIST1, .2 DIST2
203.0.113.0/30	EDGE1 – ISP	255.255.255.252	2	203.0.113.1	.2 EDGE1, .1 ISP

The /24s are summarisable as 10.10.0.0/16, which is exactly what EDGE1's NAT ACL matches. Host ranges assume no DHCP exclusions; reserve .4–.20 in each VLAN for printers and servers if DHCP is added later.

Interface address table

Device	Interface	Address	Description
EDGE1	Gi0/0	10.255.1.2/30	Inside to DIST1
EDGE1	Gi0/1	10.255.2.2/30	Inside to DIST2
EDGE1	Gi0/2	203.0.113.2/30	Outside to ISP
DIST1	Vlan10	10.10.10.2/24	HR SVI, HSRP group 10
DIST1	Vlan20	10.10.20.2/24	Finance SVI, HSRP group 20
DIST1	Vlan30	10.10.30.2/24	Engineering SVI, HSRP group 30
DIST1	Vlan40	10.10.40.2/24	Guest SVI, HSRP group 40
DIST1	Vlan99	10.10.99.11/24	Management SVI, HSRP group 99
DIST1	Gi1/0/47	10.255.10.1/30	L3 link to DIST2
DIST1	Gi1/0/48	10.255.1.1/30	L3 link to EDGE1
DIST2	Vlan10	10.10.10.3/24	HR SVI, HSRP group 10
DIST2	Vlan20	10.10.20.3/24	Finance SVI, HSRP group 20
DIST2	Vlan30	10.10.30.3/24	Engineering SVI, HSRP group 30
DIST2	Vlan40	10.10.40.3/24	Guest SVI, HSRP group 40
DIST2	Vlan99	10.10.99.12/24	Management SVI, HSRP group 99
DIST2	Gi1/0/47	10.255.10.2/30	L3 link to DIST1
DIST2	Gi1/0/48	10.255.2.1/30	L3 link to EDGE1
ACCESS1	Vlan99	10.10.99.21/24	Management, default gateway 10.10.99.1
ACCESS2	Vlan99	10.10.99.22/24	Management, default gateway 10.10.99.1

HSRP virtual addresses

VLAN	HSRP group	VIP	Virtual MAC (HSRPv1)
10	10	10.10.10.1	0000.0c07.ac0a

VLAN	HSRP group	VIP	Virtual MAC (HSRPv1)
20	20	10.10.20.1	0000.0c07.ac14
30	30	10.10.30.1	0000.0c07.ac1e
40	40	10.10.40.1	0000.0c07.ac28
99	99	10.10.99.1	0000.0c07.ac63

6. Spanning tree design

Rapid-PVST runs on all four switches with DIST1 as root for every VLAN and DIST2 as backup root, so the one blocked port in the topology always sits on DIST2.

Root placement

Switch	Command	Resulting bridge priority (typical)	Role
DIST1	`spanning-tree vlan 10,20,30,40,99 root primary`	24576 + VLAN ID	Root bridge, VLANs 10, 20, 30, 40, 99
DIST2	`spanning-tree vlan 10,20,30,40,99 root secondary`	28672 + VLAN ID	Backup root
ACCESS1/2	default	32768 + VLAN ID	Non-root

`root primary` is a macro: it sets priority to 24576, or lower if another bridge already claims a lower value. Hard-coding `spanning-tree vlan 10,20,30,40,99 priority 4096` (DIST1) and `8192` (DIST2) is more deterministic in production.

Expected port roles per VLAN

With default Gigabit path cost 4 (short method):

Switch	Port	Toward	Role	State
DIST1	Gi1/0/1, Gi1/0/2	ACCESS1, ACCESS2	Designated	Forwarding

Switch	Port	Toward	Role	State
ACCESS1	Gi1/0/1	DIST1	Root (cost 4)	Forwarding
ACCESS1	Gi1/0/2	DIST2	Designated	Forwarding
ACCESS2	Gi1/0/1	DIST1	Root (cost 4)	Forwarding
ACCESS2	Gi1/0/2	DIST2	Designated	Forwarding
DIST2	One of Gi1/0/1 or Gi1/0/2	ACCESS with lower bridge ID	Root (cost 8)	Forwarding
DIST2	The other	ACCESS with higher bridge ID	Alternate	Discarding

Which DIST2 port blocks depends on the access switches' MAC addresses (lower sender bridge ID wins the tie). Verify with `show spanning-tree vlan 10` on DIST2.

Edge-port protection

- **PortFast** on every user port (Gi1/0/10–14, 20–24, 30–34, 40–44) moves hosts straight to forwarding and suppresses topology-change floods when a PC reboots.
- **BPDU Guard** on the same ports err-disables a port that receives a BPDU, so a rogue or looped switch on a desk cannot join the tree.
- Recover an err-disabled port with `shutdown` / `no shutdown`, or add `errdisable recovery cause bpduguard` with an interval.

Interaction with HSRP

STP root is DIST1 for all VLANs, but HSRP makes DIST2 active for VLANs 20 and 40. For hosts on the access switch whose DIST2 uplink is blocking, Finance and Guest frames take ACCESS → DIST1 → other ACCESS → DIST2 to reach their gateway. It works, but adds two hops and loads the other access switch. Section 14 recommends aligning STP root with HSRP active per VLAN.

7. HSRP gateway redundancy

Each VLAN has one HSRP group whose number matches the VLAN ID; DIST1 is active for VLANs 10, 30 and 99, DIST2 for VLANs 20 and 40, and both switches lower their priority by 20 when their edge uplink (Gi1/0/48) goes down.

Group settings

VLAN	Group	VIP	DIST1 priority	DIST2 priority	Active	Standby	Preempt	Tracks Gi1/0/48 (−20)
10 HR	10	10.10.10.1	110	90	DIST1	DIST2	Both	Yes
20 Finance	20	10.10.20.1	90	110	DIST2	DIST1	Both	Yes
30 Engineering	30	10.10.30.1	110	90	DIST1	DIST2	Both	Yes
40 Guest	40	10.10.40.1	90	110	DIST2	DIST1	Both	Yes
99 Management	99	10.10.99.1	110	90	DIST1	DIST2	Both	No

HSRP version 1 and default timers (hello 3 s, hold 10 s) apply because neither is set. Splitting active roles across the pair shares routing load between the two distribution switches.

Election rules

1. Highest priority wins (default 100).
2. On a priority tie, the highest interface IP wins: DIST2 (.3) beats DIST1 (.2).
3. `preempt` lets a router take over when it would win that comparison (priority, then IP). DIST1 can never win a tie, because its addresses are always the lower ones.

Tracking math on an uplink failure

VLAN	Failing uplink	Priorities after decrement	Tie-break	Outcome
10, 30	DIST1 Gi1/0/48	DIST1 90, DIST2 90	DIST2 has the higher IP	**Works:** DIST2 preempts and becomes active

VLAN	Failing uplink	Priorities after decrement	Tie-break	Outcome
20, 40	DIST2 Gi1/0/48	DIST1 90, DIST2 90	DIST2 still has the higher IP	**Fails:** DIST2 stays active with no internet path; Finance and Guest lose internet
99	DIST1 Gi1/0/48	Not tracked: DIST1 110, DIST2 90	n/a	DIST1 stays active; management traffic bound for the internet is dropped

The fix is a decrement larger than the 20-point gap, for example `standby 20 track GigabitEthernet1/0/48 30` on both switches, so the failed router falls to 80 and its peer (90) preempts regardless of IP. See section 14, finding F1.

Sample SVI (DIST1, VLAN 10)

```
interface Vlan10
 ip address 10.10.10.2 255.255.255.0
 standby 10 ip 10.10.10.1
 standby 10 priority 110
 standby 10 preempt
 standby 10 track GigabitEthernet1/0/48 20
 ip access-group HR_IN in
```

8. Routing, internet edge and NAT

Routing is entirely static: each distribution switch sends a default route to EDGE1, EDGE1 sends a default route to the ISP, and EDGE1 returns each VLAN to the distribution switch that is HSRP-active for it, so traffic is symmetric in normal operation.

Route tables (static entries)

Device	Prefix	Next hop	Exit	Purpose
DIST1	0.0.0.0/0	10.255.1.2	Gi1/0/48	Internet via EDGE1
DIST2	0.0.0.0/0	10.255.2.2	Gi1/0/48	Internet via EDGE1
EDGE1	0.0.0.0/0	203.0.113.1	Gi0/2	Internet via ISP

Device	Prefix	Next hop	Exit	Purpose
EDGE1	10.10.10.0/24	10.255.1.1	Gi0/0	HR return via DIST1
EDGE1	10.10.20.0/24	10.255.2.1	Gi0/1	Finance return via DIST2
EDGE1	10.10.30.0/24	10.255.1.1	Gi0/0	Engineering return via DIST1
EDGE1	10.10.40.0/24	10.255.2.1	Gi0/1	Guest return via DIST2
EDGE1	10.10.99.0/24	10.255.1.1	Gi0/0	Management return via DIST1

Inter-VLAN traffic never leaves the distribution layer: both switches have all five SVIs as connected routes (`ip routing` is enabled on both). The DIST1–DIST2 routed link (10.255.10.0/30) is addressed but carries no routes, so today it is unused.

NAT / PAT

```
ip access-list standard NAT_INSIDE
 permit 10.10.0.0 0.0.255.255

interface GigabitEthernet0/0
 ip nat inside
interface GigabitEthernet0/1
 ip nat inside
interface GigabitEthernet0/2
 ip nat outside

ip nat inside source list NAT_INSIDE interface GigabitEthernet0/2 overload
```

- **Type:** Port Address Translation (NAT overload). All campus hosts share EDGE1's single public address, 203.0.113.2, distinguished by source port.
- **What is translated:** any source in 10.10.0.0/16, which covers all five VLANs including management. Transit addresses (10.255.x.x) are not translated, so pings sourced from the distribution switches' uplinks will not reach the internet; source tests from an SVI or a host.
- **Capacity:** roughly 64,000 concurrent translations per protocol per public address, ample for a mid-size office.

Packet walk: Finance host to the internet

1. Host 10.10.20.50 sends to its gateway 10.10.20.1; the HSRP virtual MAC 0000.0c07.ac14 is owned by DIST2.
2. DIST2 applies FIN_IN inbound on Vlan20 (destination is not 10.10.30.0/24, so it is permitted).
3. DIST2 follows its default route out Gi1/0/48 to 10.255.2.2.
4. EDGE1 receives on Gi0/1 (NAT inside), matches NAT_INSIDE, rewrites the source to 203.0.113.2:<port> and forwards to 203.0.113.1.
5. The reply arrives on Gi0/2, is untranslated back to 10.10.20.50, and the static route sends it to DIST2, which delivers it on VLAN 20.

Limits of the static design

If a distribution uplink fails, EDGE1 keeps sending return traffic for that switch's VLANs down the dead link (the static route is withdrawn only if EDGE1's own interface goes down), and the distribution switch has no route via its peer. Section 13 walks through the impact and section 14 proposes OSPF or floating statics.

9. Security policy and ACLs

Three extended ACLs are applied inbound on the distribution SVIs: Guest may reach only non-private destinations, Finance may not reach Engineering, and HR is permitted everywhere; the same ACLs exist on both DIST1 and DIST2 so policy holds whichever switch is HSRP-active.

ACL definitions

```
ip access-list extended GUEST_IN
 deny   ip 10.10.40.0 0.0.0.255 10.0.0.0 0.255.255.255
 deny   ip 10.10.40.0 0.0.0.255 172.16.0.0 0.15.255.255
 deny   ip 10.10.40.0 0.0.0.255 192.168.0.0 0.0.255.255
 permit ip 10.10.40.0 0.0.0.255 any

ip access-list extended FIN_IN
 deny   ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
 permit ip 10.10.20.0 0.0.0.255 any

ip access-list extended HR_IN
 permit ip 10.10.10.0 0.0.0.255 any
```

ACL	Applied to	Direction	Intent	Effect beyond the intent
GUEST_IN	Vlan40 on DIST1 and DIST2	In	Internet only; no RFC 1918 destinations	Also blocks guests from pinging their own gateway and from any internal DNS or DHCP server; anti-spoofing: sources outside 10.10.40.0/24 hit the implicit deny
FIN_IN	Vlan20 on DIST1 and DIST2	In	Finance cannot reach Engineering	Engineering-initiated sessions to Finance also fail, because Finance's replies are denied
HR_IN	Vlan10 on DIST1 and DIST2	In	Baseline placeholder	Functions only as source anti-spoofing (non-HR sources are implicitly denied)
none	Vlan30, Vlan99	n/a	Engineering and management unrestricted	Any internal VLAN can reach switch management addresses

Effective reachability matrix

Rows are the source VLAN; "Blocked (reply)" means the request leaves but the answer is dropped, so the session cannot complete.

From \ To	HR	Finance	Engineering	Guest	Management	Internet
HR	n/a	Allowed	Allowed	Blocked (reply)	Allowed	Allowed
Finance	Allowed	n/a	Blocked	Blocked (reply)	Allowed	Allowed
Engineering	Allowed	Blocked (reply)	n/a	Blocked (reply)	Allowed	Allowed
Guest	Blocked	Blocked	Blocked	n/a	Blocked	Allowed
Management	Allowed	Allowed	Allowed	Blocked (reply)	n/a	Allowed

Layer 2 controls

- Native VLAN 999 and static trunks with DTP disabled (section 4)
- BPDU Guard on every user port (section 6)
- Not yet configured: port security, DHCP snooping, Dynamic ARP Inspection, IP Source Guard, storm control, shutdown of unused ports

Device-access controls

None are configured in v1.0: there is no `enable secret`, local user, SSH, VTY access-class, AAA, banner or password encryption. This is acceptable in a lab and must be added before any real deployment (section 14, finding F6).

10. Device configuration reference

Each config file in `configs/` is a complete paste-in configuration ending in `end`; DIST1/DIST2 and ACCESS1/ACCESS2 are mirror pairs that differ only in addresses, HSRP priorities and STP root role.

EDGE1 (`configs/EDGE1.txt`)

Interface	Description	Address	NAT role
Gi0/0	INSIDE to DIST1	10.255.1.2/30	inside
Gi0/1	INSIDE to DIST2	10.255.2.2/30	inside

Interface	Description	Address	NAT role
Gi0/2	OUTSIDE to ISP	203.0.113.2/30	outside

Also configures: `no ip domain-lookup`, one default route, five static return routes, standard ACL `NAT_INSIDE`, PAT overload on Gi0/2.

DIST1 / DIST2 (`configs/DIST1.txt`, `configs/DIST2.txt`)

Port	Description	Mode	DIST1	DIST2
Gi1/0/1	TRUNK TO ACCESS1	802.1Q trunk	yes	yes
Gi1/0/2	TRUNK TO ACCESS2	802.1Q trunk	yes	yes
Gi1/0/47	L3 LINK TO peer	Routed	10.255.10.1/30	10.255.10.2/30
Gi1/0/48	L3 LINK TO EDGE1	Routed	10.255.1.1/30	10.255.2.1/30
Vlan10	HR SVI	SVI	.2, HSRP 110	.3, HSRP 90
Vlan20	Finance SVI	SVI	.2, HSRP 90	.3, HSRP 110
Vlan30	Engineering SVI	SVI	.2, HSRP 110	.3, HSRP 90
Vlan40	Guest SVI	SVI	.2, HSRP 90	.3, HSRP 110
Vlan99	Management SVI	SVI	.11, HSRP 110	.12, HSRP 90

Global setting	DIST1	DIST2
`ip routing`	enabled	enabled
VTP mode	server	server
STP	root primary, VLANs 10–99	root secondary, VLANs 10–99
Default route	10.255.1.2	10.255.2.2
ACLs	GUEST_IN, FIN_IN, HR_IN	identical

ACCESS1 / ACCESS2 (`configs/ACCESS1.txt` , `configs/ACCESS2.txt`)

Port(s)	Description	Mode	VLAN	STP features
Gi1/0/1	TRUNK TO DIST1	Trunk	native 999, allowed 10,20,30,40,99,999	n/a
Gi1/0/2	TRUNK TO DIST2	Trunk	native 999, allowed 10,20,30,40,99,999	n/a
Gi1/0/10 – 14	RAMP-HR USERS	Access	10	PortFast, BPDU Guard
Gi1/0/20 – 24	RAMP-FIN USERS	Access	20	PortFast, BPDU Guard
Gi1/0/30 – 34	RAMP-ENG USERS	Access	30	PortFast, BPDU Guard
Gi1/0/40 – 44	RAMP-GUEST APs	Access	40	PortFast, BPDU Guard
Gi1/0/3–9, 15–19, 25–29, 35–39, 45–48	unused	default (dynamic, VLAN 1, enabled)	1	none
Vlan99	Management	SVI	10.10.99.21 (ACCESS1) / .22 (ACCESS2)	n/a

Each access switch provides 5 ports per user VLAN (20 user ports); `ip default-gateway 10.10.99.1` points management traffic at the HSRP VIP. The access switches have no `ip routing` , so they act purely as Layer 2 devices with one host address for management.

11. Lab deployment guide

The lab builds in about 30 minutes: cable five devices, paste the configs in the order below (core outward), add test hosts, then run the validation in section 12.

What you need

Item	Packet Tracer	GNS3 / EVE-NG
EDGE1	ISR 4321 or 2911 (rename ports to match, or edit the config)	IOSv router
DIST1, DIST2	3650-24PS (Gi1/0/x naming, supports `ip routing` and HSRP)	IOSvL2 (uses Gi0/x naming; edit configs)
ACCESS1, ACCESS2	3650-24PS, or 2960 with port names and the encapsulation line edited	IOSvL2
ISP stand-in	One router with Gi0/0 = 203.0.113.1/30 and a loopback such as 8.8.8.8/32	Same
Test hosts	One PC per VLAN per access switch (at least HR, Finance, Engineering, Guest)	VPCS nodes

The configs use `Gi1/0/x` for switches and `Gi0/0–2` for the router. If your platform names ports differently, find-and-replace before pasting; mismatched names are the most common cause of a non-working lab.

Build steps

1. **Cable** per the link table in section 3: four trunks, two edge transits, one inter-distribution link, one ISP link.
2. **Configure the ISP stand-in:** `interface Gi0/0`, `ip address 203.0.113.1 255.255.255.252`, `no shut`, plus a loopback to ping. It needs no route back, because PAT presents everything as 203.0.113.2.
3. **Paste EDGE1:** enter `configure terminal`, paste `configs/EDGE1.txt`.
4. **Paste DIST1, then DIST2.** On physical Gi1/0/47 and Gi1/0/48 add `no shutdown` if your image leaves routed ports down.
5. **Paste ACCESS1 and ACCESS2.** After about 30 seconds `show vlan brief` should list VLANs 10–99 and 999, learned by VTP.
6. **Add hosts** with static addressing (there is no DHCP), for example:

Host	Port	IP / mask	Gateway
PC-HR	ACCESS1 Gi1/0/10	10.10.10.10/24	10.10.10.1
PC-FIN	ACCESS1 Gi1/0/20	10.10.20.10/24	10.10.20.1
PC-ENG	ACCESS2 Gi1/0/30	10.10.30.10/24	10.10.30.1
PC-GUEST	ACCESS2 Gi1/0/40	10.10.40.10/24	10.10.40.1

7. **Save** on every device: `copy running-config startup-config`.
8. **Validate** with section 12.

Common build problems

- `switchport trunk encapsulation dot1q` rejected: the platform only supports 802.1Q; delete the line.
- VLANs missing on access switches: VTP domain or password mismatch, or the trunk is not up. Check `show vtp status` on both ends.
- HSRP shows both switches active: the VLAN is not crossing the trunks between them; check `show interfaces trunk` and the allowed list.
- Guest cannot browse by name: expected, since no DNS is configured; test with an IP address.

12. Validation and test plan

Sixteen checks prove the build matches the design; run them in order after deployment and after any change, and record the result in the last column.

#	Area	Run on	Command / action	Expected result	Result
1	VTP	ACCESS1, ACCESS2	`show vtp status`	Domain RAMP-CAMPUS, mode client, same revision as DIST1	Not run
2	VLANs	All switches	`show vlan brief`	VLANs 10, 20, 30, 40, 99, 999 with RAMP-* names; user ports in the right VLAN	Not run

#	Area	Run on	Command / action	Expected result	Result
3	Trunks	All switches	`show interfaces trunk`	Gi1/0/1–2 trunking, 802.1q, native 999, allowed 10,20,30,40,99,999, mode "on"	Not run
4	STP root	DIST1	`show spanning-tree vlan 10` (repeat for 20, 30, 40, 99)	"This bridge is the root"	Not run
5	STP blocking	DIST2	`show spanning-tree vlan 10`	One port Root/FWD, the other Altn/BLK	Not run
6	Edge ports	ACCESS1	`show spanning-tree interface Gi1/0/10 detail`	PortFast enabled, BPDU Guard enabled	Not run
7	HSRP	DIST1, DIST2	`show standby brief`	DIST1 active for 10, 30, 99; DIST2 active for 20, 40; peer listed as standby	Not run
8	Gateway	Each PC	`ping <own VIP>`	Replies for HR, Finance, Engineering; **no reply** for Guest (GUEST_IN blocks 10/8)	Not run
9	Inter-VLAN	PC-HR	`ping 10.10.30.10`	Success	Not run
10	Finance policy	PC-FIN	`ping 10.10.30.10`	Fails (FIN_IN)	Not run
11	Finance policy	PC-ENG	`ping 10.10.20.10`	Fails (reply dropped by FIN_IN)	Not run
12	Finance allowed	PC-FIN	`ping 10.10.10.10`	Success	Not run
13	Guest policy	PC-GUEST	`ping 10.10.20.10` and `ping 10.10.99.11`	Both fail	Not run
14	Internet	PC-GUEST, PC-FIN	`ping 203.0.113.1` (or ISP loopback)	Success	Not run
15	NAT	EDGE1	`show ip nat translations` right after test 14	Inside local 10.10.x.x mapped to inside global 203.0.113.2	Not run
16	NAT counters	EDGE1	`show ip nat statistics`	Hits increasing; Gi0/0–1 inside, Gi0/2 outside	Not run

Useful supporting commands

- `show ip route` on DIST1/DIST2: connected 10.10.x.0/24 routes and the static default
- `show ip interface Vlan20 | include access list` confirms which ACL is applied inbound
- `show access-lists` shows per-line hit counts after the policy tests
- `show interfaces status err-disabled` finds ports BPDU Guard has shut
- `debug standby events` (lab only) watches HSRP state changes during failover tests

13. Failure scenarios

The design survives any single trunk or access-switch failure, but as configured any failure of a distribution switch or its edge uplink cuts internet access for that switch's VLANs, because return routing on EDGE1 is static and the HSRP decrement is too small for VLANs 20 and 40.

#	Failure	Layer 2 / STP	HSRP	Routing	User impact as configured
1	One trunk, e.g. DIST1–ACCESS1	ACCESS1's root port moves to Gi1/0/2; RSTP converges in about 1 s	Hellos still flow via the other access switch; no change	None	None after convergence
2	One access switch	Its trunks drop; rest of tree unaffected	No change	None	Only that switch's 20 users offline
3	DIST1–DIST2 routed link	None (it is not part of the L2 tree)	None	Link carries no routes	None
4	DIST1 uplink Gi1/0/48	None	VLANs 10, 30 fail over to DIST2 (tie, higher IP wins); VLAN 99 stays on DIST1 (untracked)	DIST1 loses its default route; EDGE1 loses the routes for 10.10.10/30/99 via 10.255.1.1	**HR, Engineering and Management lose internet:** outbound now leaves via DIST2 but replies have no return route; inter-VLAN traffic works

#	Failure	Layer 2 / STP	HSRP	Routing	User impact as configured
5	DIST2 uplink Gi1/0/48	None	VLANs 20, 40 tie at 90; DIST2 keeps active	DIST2 loses its default route; EDGE1 loses return routes for 10.10.20/40	**Finance and Guest lose internet**; inter-VLAN traffic works
6	DIST1 switch	DIST2 becomes root; RSTP reconverges	VLANs 10, 30, 99 move to DIST2 after the 10 s hold time	Return routes for 10.10.10/30/99 withdrawn on EDGE1	~10 s gateway outage for HR, Engineering, Mgmt, then **no internet** for them
7	DIST2 switch	Tree unchanged (DIST1 already root)	VLANs 20, 40 move to DIST1 after 10 s	Return routes for 10.10.20/40 withdrawn	~10 s gateway outage for Finance and Guest, then **no internet** for them
8	EDGE1 or ISP link	None	No change (track only sees Gi1/0/48)	Single exit lost	**All internet lost**; internal traffic unaffected. EDGE1 is a single point of failure by design
9	Loop or rogue switch on a user port	BPDU Guard err-disables the port	None	None	Only that port down

With the fixes in section 14 (findings F1 and F2), scenarios 4–7 become a brief failover with internet access preserved, and scenario 8 remains the only single point of failure.

Failover test procedure

1. Start a continuous ping from PC-FIN to 203.0.113.1 (`ping -t` on Windows hosts).
2. On DIST2 run `interface Gi1/0/48`, `shutdown`.
3. Watch `show standby brief` on both switches and note the ping result.
4. As configured, DIST2 remains active for VLAN 20 and pings fail; with the F1/F2 fixes, DIST1 takes over and pings resume within about 10 s.
5. `no shutdown` the uplink and confirm DIST2 preempts back to active.

14. Design review: findings and recommended fixes

The v1.0 design is sound for a lab, but two findings (F1, F2) break the high-availability goal and should be fixed first; the rest harden it for production.

ID	Severity	Finding	Recommended fix
F1	High	HSRP track decrement (20) only equals the 20-point priority gap, so VLANs 20 and 40 never fail over from DIST2 (tie goes to DIST2's higher IP)	Decrement by 30 on every group; also track Gi1/0/48 for VLAN 99
F2	High	Static return routes on EDGE1 and no route between DIST1 and DIST2 mean any distribution or uplink failure strands its VLANs' return traffic	Run OSPF area 0 on EDGE1, DIST1, DIST2 (transits plus SVIs passive), or add floating statics via the peer and the 10.255.10.0/30 link
F3	Medium	STP root is DIST1 for all VLANs while DIST2 is HSRP-active for 20 and 40, adding two hops for half of Finance/Guest traffic	Make DIST2 root primary for VLANs 20, 40 and DIST1 for 10, 30, 99
F4	Medium	VTP v2 with two servers can be wiped by a higher-revision switch; the VTP password is committed in plain text	VTP v3 with a primary server, or VTP transparent; keep secrets out of the repo
F5	Medium	No DHCP or DNS; GUEST_IN also blocks guests from internal DHCP/DNS	DHCP pools on the distribution pair or `ip helper-address`; give guests public DNS or permit UDP 53/67 to a specific server above the deny lines
F6	Medium	No device hardening: no enable secret, SSH, AAA, VTY ACL, logging, NTP	Baseline below; restrict VTY to VLAN 99 sources
F7	Medium	Unused access ports are in VLAN 1, enabled and in dynamic mode; no port security, DHCP snooping or DAI	`shutdown` and park unused ports in VLAN 999; add DHCP snooping + DAI once DHCP exists

ID	Severity	Finding	Recommended fix
F8	Low	Management VLAN reachable from HR, Finance and Engineering	Inbound ACL on Vlan99 or VTY access-class limited to an admin subnet
F9	Low	HR_IN permits everything, so the HR segment has no real policy	Replace with an approved-services list once HR systems are known
F10	Low	NAT_INSIDE translates the management VLAN to the internet	Exclude 10.10.99.0/24 unless devices need outbound access (NTP, updates)
F11	Low	EDGE1 has no inbound filtering on the ISP interface and is a single point of failure	Inbound ACL denying RFC 1918/bogon sources; a second edge router or ISP for real HA
F12	Low	Repo notes say each switch tracks its uplink, but VLAN 99 is untracked; VLAN 30 and 99 have no ACL, which the notes do not mention	Align `docs/stp-hsrp-design.md` and `docs/security-acls.md` with the configs

F1 fix (both distribution switches, all groups)

```
interface Vlan20
 standby 20 track GigabitEthernet1/0/48 30
interface Vlan99
 standby 99 track GigabitEthernet1/0/48 30
! repeat for groups 10, 30, 40
```

Newer IOS uses object tracking: `track 1 interface GigabitEthernet1/0/48 line-protocol` then `standby 20 track 1 decrement 30`.

F2 fix (OSPF option)

```
! DIST1 (DIST2 identical apart from router-id)
router ospf 1
 router-id 1.1.1.1
 passive-interface default
 no passive-interface GigabitEthernet1/0/47
 no passive-interface GigabitEthernet1/0/48
 network 10.10.0.0 0.0.255.255 area 0
 network 10.255.0.0 0.0.255.255 area 0
no ip route 0.0.0.0 0.0.0.0 10.255.1.2

! EDGE1
router ospf 1
 router-id 3.3.3.3
 network 10.255.0.0 0.0.255.255 area 0
 default-information originate
no ip route 10.10.10.0 255.255.255.0 10.255.1.1
! remove the other four campus statics likewise
```

EDGE1 then learns every VLAN from both switches and the switches learn the default from EDGE1, so any single distribution failure is routed around.

F3 fix

```
! DIST1
spanning-tree vlan 10,30,99 root primary
spanning-tree vlan 20,40 root secondary
! DIST2
spanning-tree vlan 20,40 root primary
spanning-tree vlan 10,30,99 root secondary
```

F6 hardening baseline (every device)

```
service password-encryption
enable secret <strong-secret>
username netadmin privilege 15 secret <strong-secret>
ip domain-name ramp.lab
crypto key generate rsa modulus 2048
ip ssh version 2
ip access-list standard MGMT_ONLY
 permit 10.10.99.0 0.0.0.255
line vty 0 15
 transport input ssh
 login local
 access-class MGMT_ONLY in
logging host <syslog-server>
ntp server <ntp-server>
banner motd ^Authorized access only^
```

15. Operations runbook and troubleshooting

Routine changes follow the same pattern: make the change on DIST1 (and its mirror on DIST2), re-run the relevant checks from section 12, save, and record it in the change log.

Add a new VLAN (example: VLAN 50 RAMP-SALES, 10.10.50.0/24)

1. On DIST1 only (VTP propagates): `vlan 50`, `name RAMP-SALES`.
2. On DIST1 and DIST2: create `interface Vlan50` with .2 / .3, `standby 50 ip 10.10.50.1`, priorities 110/90 per the load-sharing plan, `preempt`, and the uplink track.
3. On all eight trunk ports: `switchport trunk allowed vlan add 50` (never omit `add`, which would replace the list).
4. On access switches: assign ports with `switchport access vlan 50`, PortFast and BPDU Guard.
5. Add STP root commands for VLAN 50 on DIST1/DIST2.
6. On EDGE1: `ip route 10.10.50.0 255.255.255.0 <HSRP-active switch>` (not needed after F2). NAT already covers 10.10.0.0/16.
7. Update the VLAN and IP tables in this document and the repo `README.md`.

Onboard a user port

```
interface GigabitEthernet1/0/15
 description RAMP-HR USERS
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
```

Planned maintenance on a distribution switch

1. Move its HSRP active groups to the peer: lower priority, e.g. `standby 10 priority 80` on DIST1.
2. Confirm with `show standby brief` on both switches.
3. Because of F2, also move EDGE1's static return routes for those VLANs to the peer before reloading.
4. Perform the work, restore priorities and routes, and verify preemption.

Troubleshooting guide

Symptom	Likely cause	Check	Fix
Host cannot ping its gateway	Wrong access VLAN, port err-disabled, or (Guest) ACL by design	`show interfaces status`, `show vlan brief`, `show standby brief`	Correct VLAN; `shut` / `no shut` err-disabled port; Guest gateway pings are blocked intentionally
Both DIST switches show HSRP Active	Hellos not crossing the access layer	`show interfaces trunk` on all switches	Add the VLAN to the allowed list, bring trunk up
User port goes err-disabled	BPDU Guard saw a switch or loop	`show interfaces status err-disabled`, `show log`	Remove the device, then bounce the port
VLAN missing on an access switch	VTP mismatch or trunk down	`show vtp status`, `show vtp password`	Match domain/password/version

Symptom	Likely cause	Check	Fix
Internet fails for some VLANs only	Uplink down on the HSRP-active switch (F1/F2)	`show ip route 0.0.0.0` on DIST, `show ip route` on EDGE1	Restore uplink, or move HSRP active to the peer
Internet fails for everyone	EDGE1 or ISP link down; NAT misconfigured	`show ip nat statistics`, `ping 203.0.113.1` from EDGE1	Restore link; verify `ip nat inside/outside` on the right interfaces
Finance cannot reach an Engineering server	FIN_IN by design	`show access-lists FIN_IN` hit counts	Add a specific permit above the deny if the business approves
Traffic takes an extra hop through the other access switch	STP root and HSRP active differ (F3)	`show spanning-tree vlan 20` on access switches	Align root with HSRP active

Configuration backup

Keep `configs/` as the source of truth: after any change, `show running-config` on the device, strip passwords, and commit the updated file with a change-log entry.

16. Change log and glossary

Change log

Version	Changes
v1.0	VLAN segmentation and VTP domain; Rapid-PVST root at distribution; HSRP VIPs for all VLANs with load distribution; ACLs for Guest isolation and Finance segmentation; NAT overload at the edge; draw.io diagram and validation checklist

Version	Changes
v1.1 (proposed)	Fixes F1–F3 from section 14: HSRP decrement 30 and VLAN 99 tracking, OSPF between EDGE1 and distribution, per-VLAN STP root aligned with HSRP

Glossary

Term	Meaning
ACL	Access control list; ordered permit/deny rules, ending in an implicit deny
BPDU Guard	Err-disables an edge port that receives spanning-tree BPDUs
DTP	Dynamic Trunking Protocol; disabled here with `switchport nonegotiate`
HSRP	Hot Standby Router Protocol; two routers share a virtual gateway IP and MAC
Native VLAN	The VLAN whose frames cross an 802.1Q trunk untagged (999 here)
PAT / NAT overload	Many private addresses share one public address, separated by port number
PortFast	Moves an edge port straight to forwarding without STP listening/learning
Preempt	Lets a higher-priority HSRP router reclaim the active role
Rapid-PVST	Cisco per-VLAN Rapid Spanning Tree (802.1w), one tree per VLAN
RFC 1918	Private IPv4 ranges: 10/8, 172.16/12, 192.168/16
SVI	Switched virtual interface; a VLAN's routed interface on a Layer 3 switch
VIP	Virtual IP; the HSRP gateway address hosts use

Term	Meaning
VTP	VLAN Trunking Protocol; servers distribute the VLAN database to clients

Sources

- Project repository: syedjafri06193/Cisco-Enterprise-Network-Arch-proj — configs, diagram and notes as of this document's date